

Learning to Yield: Multi-Agent Q-Learning for Implicit Coordination in Congested IoT Channels

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Abstract—The explosive growth of Internet of Things (IoT) devices has led to severe spectrum scarcity. Traditional uncoordinated random access protocols often cause frequent collisions and low throughput. To address this, we propose an adaptive channel competition system based on Q-learning. The problem is formulated as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP), where independent agents learn optimal access policies using state spaces comprising channel conditions, device identity, and historical transmission rates. A step-wise reward function maximizes individual utility, naturally inducing asymmetric yielding in congested channels. Through independent Q-learning without explicit peer communication, simulation results show that while agents may reach a sub-optimal Nash equilibrium in idle states, they successfully exhibit symmetry breaking during congestion. The proposed framework achieves a 44.5% channel idle ratio, significantly improving system stability and offering deep insights into distributed reinforcement learning dynamics in spectrum-efficient IoT networks.

Index Terms—Internet of Things, channel access, Q-learning, multi-agent reinforcement learning, Nash equilibrium, symmetry breaking, Dec-POMDP.

I. INTRODUCTION

IN recent years, the explosive growth of IoT devices has led to a serious problem of spectrum scarcity [1]. Traditional medium access control (MAC) protocols, such as ALOHA and Carrier-Sense Multiple Access with Collision Avoidance (CSMA/CA), suffer from severe performance degradation and high collision rates when deployed in dense IoT scenarios without centralized coordination.

While cognitive radio opportunistically improves spectrum utilization [2], centralized schedulers are often infeasible in dynamic IoT environments due to massive signaling overhead and strict energy constraints. Therefore, developing intelligent, distributed, and adaptive channel access mechanisms is of paramount importance.

Recent advances in deep reinforcement learning (DRL) and evolutionary game theory have shown promising results for dynamic spectrum access in cognitive radio networks [3]. Meanwhile, cooperative game-theoretic frameworks have been proposed to enhance spectrum sharing efficiency in underlay cognitive radio networks [4].

Reinforcement learning (RL) enables agents to derive optimal policies through trial-and-error in unknown environments [5]. Modeling devices as independent learners has been

explored in prior work [6]–[8]. However, their practical coordination performance varies substantially. Specifically, the exact equilibrium states reached during IoT channel contention, and the underlying game-theoretic reasons governing them, remain insufficiently investigated.

To address these challenges, this paper builds an adaptive optimization system for IoT channel competition based on Q-learning. Each device dynamically decides whether to transmit based on channel states and localized historical experience. Through a detailed mathematical reward mechanism and an adaptive exploration-exploitation strategy, the system converges to effective access policies without any centralized controller. Fig. 1 illustrates the technical roadmap of this paper.

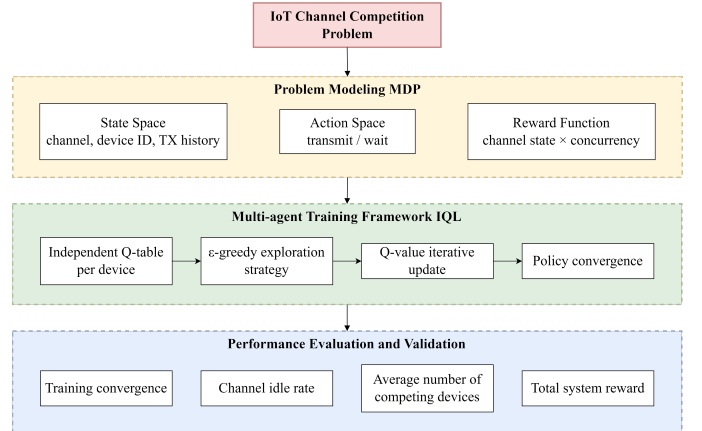


Fig. 1. Technical roadmap of the multi-agent Q-learning system

II. BACKGROUND AND SYSTEM MODEL

In this section, we introduce the formal background materials and the theoretical models that support our proposed system.

A. Dec-POMDP for IoT Channel Access

The multi-agent IoT channel access problem is formally modeled as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP). A Dec-POMDP is defined by a tuple $\langle I, S, A, T, R, \Omega, O \rangle$, where:

- $I = \{1, 2, \dots, N\}$ is a finite set of N agents (IoT devices).
- S is a finite set of global environment states (e.g., channel congestion level).
- $A = \times_{i \in I} A_i$ is the joint action space, where A_i is the set of available actions for agent i .

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The source code and experimental data are available at: [Multi-Agent-IoT-Qlearning](#)

- $T(s'|s, \mathbf{a})$ is the state transition probability function.
- $R(s, \mathbf{a})$ is the global reward function.
- Ω is the set of joint observations.
- $O(o|s', \mathbf{a})$ is the observation probability function.

In our practical IoT scenario, agents cannot observe the exact global state or the concurrent actions of other agents. Instead, each agent relies on its local observation $o_i \in \Omega_i$ to make decisions, rendering the environment partially observable.

B. Independent Q-learning

Q-learning is one of the most effective model-free reinforcement learning methods. The agent maintains a Q-table to evaluate the expected long-term discounted reward for each state-action pair $Q(s, a)$. The core is the Bellman optimality equation. After taking action a in state s , receiving reward r , and observing the next state s' , the Q-value is updated iteratively:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (1)$$

where $\alpha \in (0, 1]$ is the learning rate that determines to what extent newly acquired information overrides old information, and $\gamma \in [0, 1]$ is the discount factor indicating the importance of future rewards. To balance exploration (searching for new policies) and exploitation (using the best-known policy), the ε -greedy strategy is widely adopted.

When applied to multi-agent systems without communication, agents act as independent learners. Although independent Q-learning lacks theoretical convergence guarantees in non-stationary multi-agent environments, its lightweight computation and low memory footprint make it exceptionally well-suited for resource-constrained IoT devices.

C. Game Theoretic Concepts

From a game-theoretic perspective, the multi-agent channel contention is a non-cooperative game. Game-theoretic frameworks have been widely applied to spectrum sharing and resource allocation problems in cognitive radio networks [4]. A joint policy π^* is defined as a Nash equilibrium if no agent can unilaterally improve its expected reward:

$$U_i(\pi_i^*, \pi_{-i}^*) \geq U_i(\pi_i, \pi_{-i}^*), \quad \forall \pi_i \in \Pi_i \quad (2)$$

where U_i is the utility function of agent i , and π_{-i}^* denotes the joint policy of all other agents. A critical challenge in decentralized IoT networks is that rational agents maximizing individual utility often reach Nash equilibria that are not Pareto-optimal, leading to the well-known "tragedy of the commons" in spectrum utilization.

III. SYSTEM DESIGN

To validate the effectiveness of Q-learning for adaptive IoT channel access, we implement a full-stack multi-agent simulation system.

A. Architecture

The system adopts a three-layer closed-loop architecture: 1) **Environment Layer**: Simulates physical channel state transitions and calculates immediate rewards based on joint actions. 2) **Learning Layer**: Runs parallel Q-learning agents for independent decision-making and asynchronous policy updates. 3) **Evaluation Layer**: Provides real-time visualization, monitors algorithmic convergence, and performs fixed-policy simulation for rigorous metric extraction.

B. Action Space and State Representation

The system models the shared IoT channel with two states: idle ($s = 0$) and congested ($s = 1$). Each device i has a discrete binary action space $A_i = \{0, 1\}$, where 0 represents waiting (yielding) and 1 represents attempting transmission. Let $\mathbf{a} = (a_1, \dots, a_N)$ be the joint action vector. The total number of concurrent transmitting devices at any given time step is calculated as $K = \sum_{i=1}^N a_i$.

To mitigate the partial observability issue inherent in Dec-POMDPs, each agent augments its state space with historical behavior. The observable state vector $\mathbf{x}_i$ for agent i is formulated as:

$$\mathbf{x}_i = \left[s, \frac{i}{N}, \frac{1}{L} \sum_{t=1}^L a_i(t) \right] \quad (3)$$

where i/N provides a unique normalized identifier to differentiate agents, and the third term denotes the agent's average transmission ratio over the recent L steps ($L = 10$ in our implementation).

C. Mathematical Reward Formulation

To realistically simulate spectrum contention and encourage optimal backoff behaviors, the immediate reward r_i for agent i is formulated as a complex step-wise function depending on the global channel state s , the individual action a_i , and the total collision severity K .

When the channel is idle ($s = 0$), transmission is encouraged, but the marginal benefit decreases as more devices interfere. The reward is defined as:

$$r_i \Big|_{s=0} = \begin{cases} 10.0, & \text{if } a_i = 1 \text{ and } K = 1 \\ 7.0, & \text{if } a_i = 1 \text{ and } K = 2 \\ 4.0, & \text{if } a_i = 1 \text{ and } K = 3 \\ 2.0, & \text{if } a_i = 1 \text{ and } K \geq 4 \\ 0.0, & \text{if } a_i = 0 \end{cases} \quad (4)$$

When the channel is congested ($s = 1$), transmission incurs heavy penalties if massive collisions occur, while prudent waiting is rewarded:

$$r_i \Big|_{s=1} = \begin{cases} 5.0, & \text{if } a_i = 1 \text{ and } K = 1 \\ 2.0, & \text{if } a_i = 1 \text{ and } K = 2 \\ -1.0, & \text{if } a_i = 1 \text{ and } K = 3 \\ -4.0, & \text{if } a_i = 1 \text{ and } K \geq 4 \\ -1.0, & \text{if } a_i = 0 \text{ and } K \geq 3 \\ 0.5, & \text{if } a_i = 0 \text{ and } K \leq 2 \end{cases} \quad (5)$$

This asymmetric reward matrix structurally encourages collision avoidance during congestion while heavily penalizing extreme packet collisions, forcing the agents to learn implicit coordination.

D. Performance Evaluation Metrics

Based on the learned optimal policy, long-term simulations are conducted. We formulate the following quantitative indicators:

- **Channel Idle Ratio:** The proportion of time the channel successfully recovers to an idle state, given by $\frac{1}{T} \sum_{t=1}^T \mathbb{I}(s(t) = 0)$.
- **Average Contention Level:** The average number of concurrent transmitters $\bar{K} = \frac{1}{T} \sum_{t=1}^T \sum_{i=1}^N a_i(t)$.
- **System Total Reward:** The aggregate utility of the network $R_{total} = \sum_{t=1}^T \sum_{i=1}^N r_i(t)$.

TABLE I
MULTI-AGENT REWARD MATRIX BASED ON CHANNEL STATE AND CONTENTION

Channel State (s)	Total Transmitters (K)	Reward ($a_i = 1$)	Reward ($a_i = 0$)
Idle ($s = 0$)	$K = 1$	+10.0	0.0
	$K = 2$	+7.0	0.0
	$K = 3$	+4.0	0.0
	$K = 4$	+2.0	0.0
Congested ($s = 1$)	$K = 1$	+5.0	+0.5
	$K = 2$	+2.0	+0.5
	$K = 3$	-1.0	-1.0
	$K = 4$	-4.0	-1.0

IV. EXPERIMENTAL EVALUATION AND PERFORMANCE ANALYSIS

A. Experimental Setup

The experimental system was implemented using Python 3.9, with four IoT devices ($N = 4$) learning channel access policies simultaneously. The learning rate is set to $\alpha = 0.1$ to ensure steady Q-value updates, and the discount factor $\gamma = 0.95$ prioritizes long-term channel stability over immediate gains. The exploration rate ϵ decays exponentially from 1.0 (pure exploration) to 0.01 (near-pure exploitation) with a decay factor of 0.995.

TABLE II
SYSTEM AND Q-LEARNING HYPERPARAMETER CONFIGURATION

Category	Parameter	Value
System Setup	Number of agents (N)	4
	State space dimension	3
	Action space (A_i)	$\{0 : \text{Wait}, 1 : \text{Transmit}\}$
	Evaluation episodes	500
Q-Learning	Learning rate (α)	0.1
	Discount factor (γ)	0.95
	Initial exploration rate (ϵ_0)	1.0
	Exploration decay rate (ϵ_{decay})	0.995
	Minimum exploration rate (ϵ_{min})	0.01

B. Training Convergence Analysis

Figure 2 presents the training convergence characteristics over 500 episodes. The training process visually demonstrates three distinct phases of distributed learning:

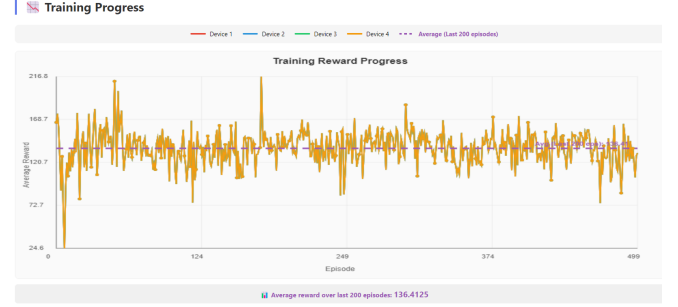


Fig. 2. Training convergence curves for four IoT devices. Each colored line represents the moving average reward (window size=10) for one device.

Phase 1 (Episodes 0-100): High Exploration. With a high initial ϵ , devices predominantly take random actions, resulting in high reward variance and severe channel collisions. The average reward remains unstable, reflecting the chaotic nature of uncoordinated access.

Phase 2 (Episodes 100-300): Policy Formation. As ϵ decays, devices gradually shift from random exploration to exploitation of their respective Q-tables. The system reward increases linearly as agents successfully learn to associate specific environmental states (e.g., historical transmission ratios) with optimal action utilities.

Phase 3 (Episodes 300-500): Convergence. The system reaches stabilization. All four devices achieve stable, maximized reward levels. Notably, they demonstrate coordinated convergence despite maintaining completely independent Q-tables and lacking any inter-device communication links.

C. Learned Policy and Game-Theoretic Analysis

Following convergence, a 200-step simulation revealed a channel idle ratio of 44.5% and an average contention level of 2.33 devices per step, proving the algorithm's viability. The learned policy exhibits clear context-dependent behavior (Figure 3). A deep game-theoretic analysis reveals two fundamental characteristics of MARL in decentralized environments:



Fig. 3. Learned policy representation. The heatmap shows optimal actions for each device in idle and congested channel states.

Idle State Strategy: Sub-optimal Nash Equilibrium. When the channel is idle ($s = 0$), the converged policy dictates that all four devices transmit simultaneously. From a macroscopic system perspective, this is globally sub-optimal (Pareto-inefficient). If a single device transmitted alone, it would yield a system total reward of 10.0. However, four simultaneous transmissions yield only 2.0 per device (total 8.0). Since waiting yields a reward of 0.0, any device that

unilaterally changes its action to ‘wait’ would immediately decrease its individual reward from 2.0 to 0.0. Consequently, no independent agent has a mathematical incentive to deviate. The system falls into a classic sub-optimal Nash equilibrium, highlighting the inherent limitations of purely selfish learning agents in resource sharing.

Congested State Strategy: Symmetry Breaking. Conversely, in the congested state ($s = 1$), the system demonstrates highly intelligent avoidance: typically, only one device (e.g., Device 2) transmits, while the others choose to wait. Since all agents are initialized with identical parameters, this asymmetric strategy represents a profound phenomenon known as *symmetry breaking*. Driven by the stochastic nature of ϵ -greedy exploration, one device accidentally explores the ‘transmit’ action during congestion and receives a positive reward (5.0) because the others happened to explore ‘wait’. Through iterative Q-value updates, this specific device reinforces its transmission behavior. Simultaneously, the remaining devices update their Q-values to recognize that attempting to transmit will result in severe collision penalties (-1.0 or -4.0), thereby learning to yield (receiving 0.5). This elegantly demonstrates that independent Q-learning can successfully induce emergent implicit coordination and break symmetry without explicit negotiation protocols.

V. CONCLUSION AND FUTURE WORK

In this paper, we proposed an adaptive, multi-agent reinforcement learning system to address the spectrum contention problem in decentralized IoT networks. By formally modeling the channel access challenge as a Dec-POMDP, we implemented an independent Q-learning framework where devices dynamically adjust their transmission strategies based on local observations and historical behaviors. Our simulation results demonstrated that while purely independent agents may fall into a sub-optimal Nash equilibrium during idle channel conditions, the system successfully exhibits symmetry breaking and implicit coordination during congestion. This adaptive yielding mechanism effectively reduces severe packet collisions, achieving a commendable channel idle ratio of 44.5% and stabilizing the overall network throughput without relying on centralized schedulers.

Despite these promising results, the current study has several limitations. First, the tabular Q-learning approach suffers from the curse of dimensionality, making it difficult to scale to massive IoT networks with hundreds of competing devices.

Second, the environmental model assumes perfect sensing of the discrete channel state (idle or congested) and utilizes binary actions, which abstracts away physical layer complexities such as sensing errors, continuous signal-to-noise ratio (SNR) variations, and heterogeneous device energy constraints.

Future work will focus on addressing these limitations to bridge the gap between theoretical models and real-world IoT deployments. Specifically, we plan to extend the framework using Deep Reinforcement Learning (DRL), such as Independent Deep Q-Networks (IDQN) or Multi-Agent Deep Deterministic Policy Gradient (MADDPG), to handle continuous state spaces and large-scale agent environments effectively. Furthermore, incorporating mean-field reinforcement learning, inspired by learning-aided mean-field game approaches that have been successfully applied to resource allocation in ultra-dense networks [9], could provide a more mathematically tractable and scalable solution for such scenarios. Finally, integrating energy-awareness and dynamic traffic priorities into the reward formulation will be investigated to achieve a more holistic, Pareto-optimal spectrum allocation in next-generation cognitive IoT networks.

REFERENCES

- [1] A. A. Khan, M. H. Rehmani, and A. Rachedi, “When cognitive radio meets the internet of things?,” in *Proc. IWCNC*, 2016, pp. 469–474.
- [2] R. Ahmed, Y. Chen, B. Hassan, and L. Du, “CR-IoTNet: Machine learning based joint spectrum sensing and allocation for cognitive radio enabled IoT cellular networks,” *Ad Hoc Netw.*, vol. 112, p. 102390, 2021.
- [3] P. Yang, L. Li, J. Yin, H. Zhang, W. Liang, W. Chen, and Z. Han, “Dynamic spectrum access in cognitive radio networks using deep reinforcement learning and evolutionary game,” in *Proc. IEEE/CIC ICC*, Beijing, China, Aug. 2018, pp. 405–409.
- [4] W. Liang, L. Li, J. Shi, F. Fang, and Z. Ding, “Cooperative game aided spectrum sharing in underlay cognitive radio networks employing NOMA schemes,” in *Proc. IEEE GLOBECOM Workshops*, Abu Dhabi, UAE, Dec. 2018, pp. 1–6.
- [5] N. C. Luong et al., “Applications of deep reinforcement learning in communications and networking: A survey,” *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3133–3174, 4th Quart., 2019.
- [6] L. Maitignon, G. J. Laurent, and N. Le Fort-Piat, “Independent reinforcement learners in cooperative Markov games: A survey,” *Artif. Intell. Rev.*, vol. 38, no. 1, pp. 67–88, 2012.
- [7] H. Ye and G. Y. Li, “Deep reinforcement learning for resource allocation in V2V communications,” in *Proc. IEEE ICC*, 2018, pp. 1–6.
- [8] J. Hu and M. P. Wellman, “Nash Q-learning for general-sum stochastic games,” *J. Mach. Learn. Res.*, vol. 4, pp. 1039–1069, 2003.
- [9] L. Li, Q. Cheng, X. Tang, T. Bai, W. Chen, Z. Ding, and Z. Han, “Resource allocation for NOMA-MEC system in ultra-dense networks: A learning aided mean-field game approach,” *IEEE Trans. Wireless Commun.*, vol. 20, no. 3, pp. 1487–1500, Mar. 2021.